

Oswald Gyabaah

+233 55 413 4307 | oswald.gyabaah@gmail.com | linkedin.com/in/ogyabaah | github.com/cyteger | Accra, Ghana (UTC+0)

PROFESSIONAL SUMMARY

Hands-on solution architect and engineering leader who designs and builds distributed, event-driven systems for regulated finance - and still writes the code. Founding CTO of MyHealthCop, where I architected the full platform on AWS, and founder of Virk Cloud, a production Kubernetes PaaS built from scratch. Now run the on-prem core banking platform behind 1.5M+ daily financial transactions under bank regulatory scrutiny. I set security-first, cloud-native standards, build production-grade proofs of concept, and integrate applied AI - LLM agents, RAG, tool-use - into engineering workflows.

CORE COMPETENCIES

Solution Architecture

Distributed & Event-Driven Systems

Microservices & Cloud-Native Design

Regulated Fintech & Banking

Security-First Architecture

AWS & GCP

CI/CD & Automated Testing

Applied AI (LLMs, Agents)

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD

May 2025 - Present

Core Banking Services Lead - Accra, Ghana

- Own the on-prem Finacle core banking platform sustaining **1.5M+ daily financial transactions** under bank regulatory scrutiny, across 800+ production pods on Tanzu Kubernetes Grid.
- Architected integrations between Finacle core banking and third-party systems - transaction portals and internal tools - designing service boundaries and API contracts.
- Designed and built a bank-wide OTP identity service (Python/Django + Next.js) securing core banking access for 600+ users, integrating SMS and email gateways with structured audit logging.
- Designed a national ID verification (KYC) service with automated document processing and self-hosted MinIO storage - a secure-by-default service in a regulated environment.

Guaranty Trust Bank Ghana LTD

Apr 2024 - Present

DevOps Engineering Lead - Accra, Ghana

- Founded and led the bank's DevOps practice from zero - a 4-engineer team - driving adoption of Git workflows, containerized Kubernetes deployments, and GitOps through influence and demonstrated value, not mandate.
- Architected high-availability Kubernetes clusters across multiple on-prem environments running internal applications at 3M+ daily requests with 99.97% measured uptime, with auto-scaling and self-healing.
- Implemented CI/CD automation that cut deployment time ~90% and increased release velocity ~5x; introduced GitOps with FluxCD; integrated Veracode and Trivy security scanning into the pipeline.
- Introduced a unified observability stack (Prometheus, Grafana, Elastic Stack, Sentry) for centralized logging, tracing, and real-time alerting, reducing mean time to resolution on production incidents.

MyHealthCop LTD

Jan 2021 - Mar 2024

Chief Technical Officer (Founding) - Accra, Ghana

- Architected the myhealthcop.com platform end to end - system design, service boundaries, database modeling, and API design - and led a 12-person cross-functional engineering team through the full delivery lifecycle.
- Designed secure AWS infrastructure running containerized microservices on EKS and ECS alongside Lambda for event-driven workloads, with autoscaling, load balancing, and inter-service communication.
- Built a high-throughput event-driven processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS, with multi-provider failover for SMS, email, and payments.
- Set the engineering quality bar - review norms, testing standards, and CI/CD on GitHub Actions and Terraform - making deployments repeatable, auditable, and reversible.
- Designed a disaster recovery strategy mirroring the AWS production environment on GCP for rapid failover, with secure VPCs, IAM policies, and dual-environment CI/CD.

PROJECTS

Virk Cloud (virk.cloud) - Kubernetes PaaS Architecture

Founder & CTO. Architected and built a production Platform-as-a-Service from scratch - instant cluster deployment with built-in CI/CD, observability, and log streaming. Built the management console for workload management, custom domains, and monitoring; manage the underlying bare-metal infrastructure end to end.

Stack: Kubernetes (RKE2), FluxCD, Terraform, Next.js, TypeScript, Convex, Prometheus, Grafana, Python

Digital-Recon - Transaction Reconciliation Engine Event-Driven

Founder. Real-time transaction reconciliation engine for financial institutions, detecting discrepancies across heterogeneous data sources - relational databases, APIs, Elasticsearch. Architected an event-driven processing layer using Kafka for upstream transaction ingestion and Celery for scheduled and on-demand reconciliation jobs.

Stack: Python, Celery, Kafka, Elasticsearch, PostgreSQL, React, TypeScript, Docker

AI Agents for Engineering Operations Applied AI

Personal R&D. Building specialized AI agents for QA automation, code review, and sysadmin operations, running against live Linux production servers. Prototyping agent-driven tooling with RAG, tool-use, and multi-agent orchestration to integrate AI responsibly across the SDLC.

Stack: Python, Anthropic API, OpenAI API, RAG, tool-use, AWS Lambda, EventBridge

EDUCATION

BSc Computer Engineering - Ashesi University Sep 2016 - May 2020

Berekuso, Ghana

CERTIFICATIONS

AI Engineering Specialization - ByteByteAI	Mar 2026
Oracle Certified Architect Associate - Oracle	Aug 2025
Oracle Certified DevOps Professional - Oracle	Aug 2025
AWS Certified Cloud Practitioner - AWS	May 2024

SKILLS

- Architecture & Design:** Solution architecture, microservices, event-driven architecture, distributed systems, cloud-native & 12-factor design, system design, API design, database modeling
- Backend:** Python (Django, FastAPI, Celery), Node.js (Express), REST APIs, GraphQL, WebSockets
- Cloud & Infrastructure:** AWS (EKS, ECS, Lambda, SQS/SNS, EventBridge), GCP (GKE, Cloud SQL), Kubernetes, Docker, Helm, Terraform
- Messaging & Events:** Kafka, AWS SQS/SNS, EventBridge, RabbitMQ, Redis queues
- CI/CD & Testing:** GitHub Actions, Azure DevOps, FluxCD, ArgoCD; Jest, React Testing Library, Cypress, Playwright E2E, pytest
- Security:** OAuth2/Keycloak, OIDC, JWT, LDAP, WAF, Veracode, Trivy, KYC/identity verification
- Applied AI:** LLM integration (OpenAI, Anthropic), RAG, embeddings, AI agents, multi-agent orchestration, tool-use patterns
- Observability:** Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Sentry